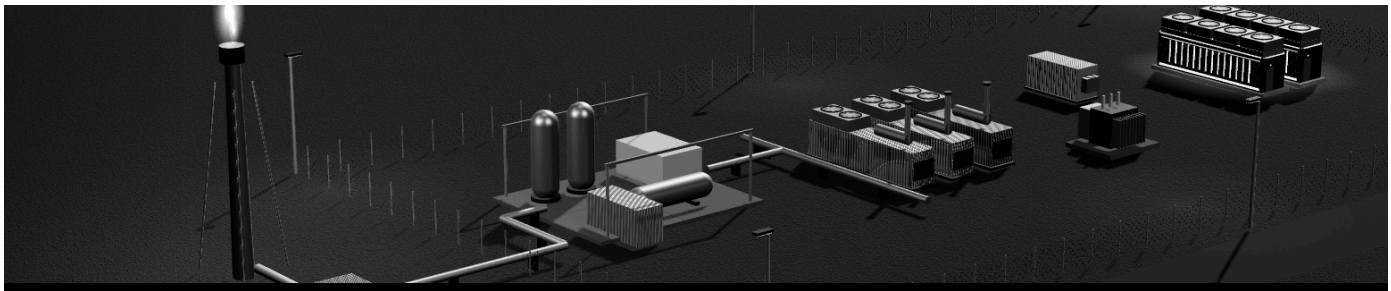
Cowlitz County Headquarters Landfill, Castle Rock, Washington: a first sizing

What the gas the site now flares could power, how much land it would take, and what it could earn each year.

2.0 MW	2	7 ac	187k
POWER FROM THE GAS	COMPUTE MODULES	LAND NEEDED	MMBTU BOUGHT / YR

GAS SALE, PER YEAR	LAND LEASE, PER YEAR	TOTAL TO THE OWNER, PER YEAR
up to $140k	$7k to $15k	$7k to $155k



YOUR SITE IN OUR LIVE 3D MODEL — ACCRETIONCOMPUTE.COM/SITE-MODEL?SRC=LFG&FLOW=769&CH4=48

THE MATH

Landfill gas collected and flared (2022)	769 scfm
Methane content (2022)	48%
Energy in the gas	22.4 MMBtu per hour
Power at a heat rate of 11,000 Btu/kWh	about 2.0 MW
Compute (power less cooling and losses, /1.2)	about 1.7 MW
Compute modules (about 1 MW each)	2
Gas engines (running plus one spare)	3 at 1.6 MW
Land: 5 acres plus 1.2 acres per MW	about 7 acres
Gas bought per year (95% run time)	187k MMBtu
Gas payment, up to $0.75/MMBtu	up to $140k
Land lease at $1,000 to $2,000 per acre per year	$7k to $15k

Landfill gas: about 1,012 Btu per cubic foot of methane. Prices in US dollars; the price band is the one shown on our Site model page.

WHAT IT WOULD LOOK LIKE

- Gas engines on a small parcel near the gas, a battery to smooth the load, and factory-built compute modules leased to an AI company.
- No grid connection and no export: all of the power is used on site.
- We build and run the site.
- Multi-year gas and land agreements, matched to the tenant's lease, at a fixed price.

WORTH CHECKING

- Where the County's 2026 gas management RFP stands
- Washington emission rules for engines
- Space for a small fenced parcel near the flare
- Gas quality (siloxanes, H2S) for engines

Source: EPA Landfill Methane Outreach Program (LMOP) database, September 2024 release (2022 data): gas collected and flared; the earlier engine plant is suspended. This is a first read from public data, not an offer: no cost and no obligation. The real price and terms are set in the gas and land agreements after a site review.